

Research on Intrinsic Motivation for Reinforcement Learning

Correlation-Aware Reward Exploration (CARE)

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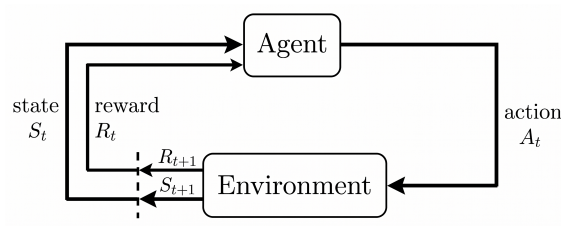
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Bachelor's Thesis – Faculty of Information Technology

Reinforcement Learning and Sparse Reward

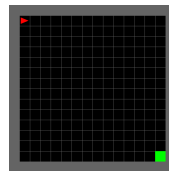


Dense Reward



MuJoCo Humanoid

Sparse Reward



MiniGrid Empty-16x16

Three Sources of Intrinsic Reward

Novelty

*"Have I been
here before?"*

Count-based

Surprise

*"Did I
predict that?"*

ICM

Influence

*"Did I change
the world?"*

RIDE

Shaped reward: $r_t = r_t^{\text{ext}} + \beta r_t^{\text{int}}$

- Same β for every state – but useful exploration varies per state.
- Static throughout training – exploration need shifts as the agent learns.
- Decoupled from task signal – intrinsic bonus fires whether or not it helps extrinsic reward.

Research question

Can we adapt the intrinsic-reward coefficient automatically during training so that exploration better supports extrinsic-reward learning?

Before: $r_t = r_t^{\text{ext}} + \beta r_t^{\text{int}}$

CARE: $r_t = r_t^{\text{ext}} + \boxed{\beta_\psi(s_t)} r_t^{\text{int}}$

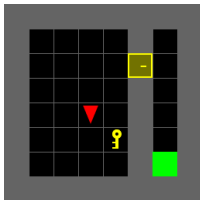
Main idea

Raise intrinsic reward when exploration helps future extrinsic reward; reduce it when distracting.

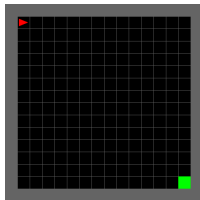
- **Correlation** – align intrinsic reward with future extrinsic advantage.
- **Regularizer** – keep β_ψ near β_0 in log-space.

$$\mathcal{L}_\beta(\psi) = \underbrace{-\mathbb{E}[\hat{I}_z \cdot \hat{A}_z^E]}_{\text{correlation}} + \underbrace{\lambda_{\text{reg}} (\log \beta_\psi(s) - \log \beta_0)^2}_{\text{log-space regularizer}}$$

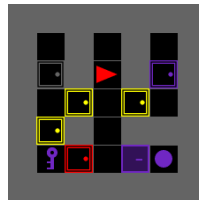
Experimental Design



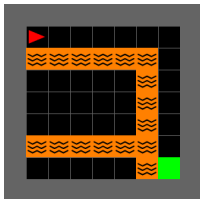
DoorKey-8x8



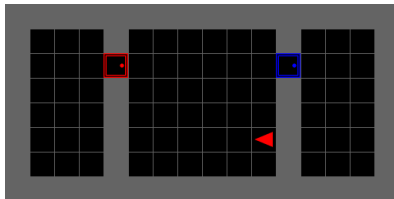
Empty-16x16



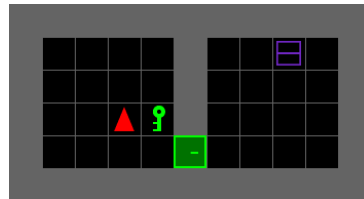
KeyCorridor-S3R3



LavaCrossing-S9N3



RedBlueDoors-8x8

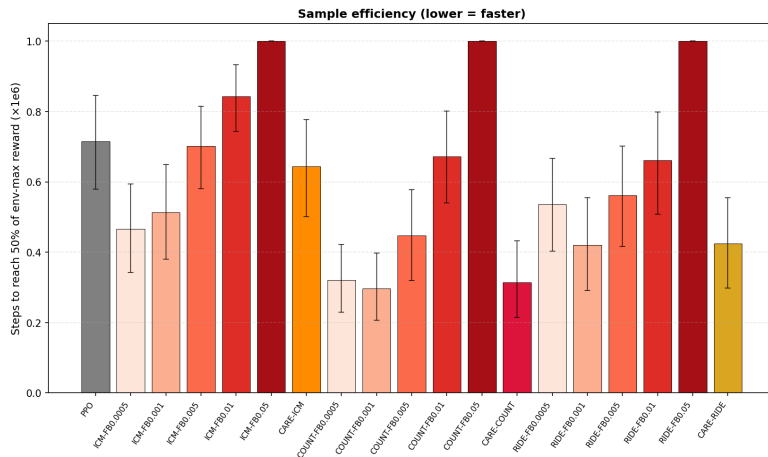


UnlockPickup

All sparse-reward, discrete grid. Red triangle = agent.

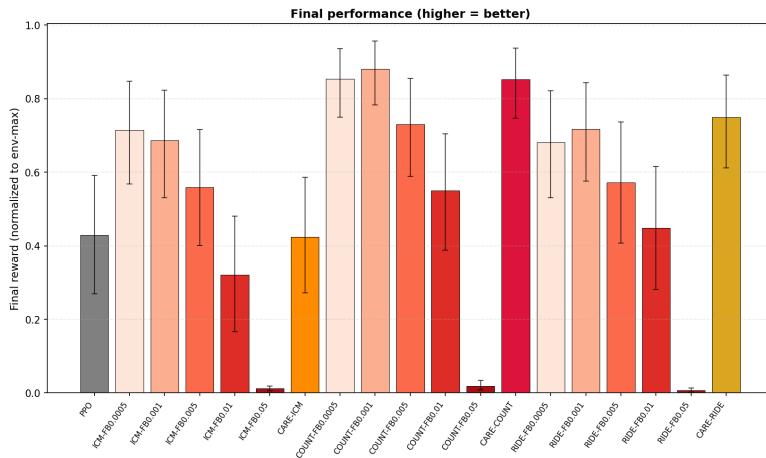
Base algorithm	PPO (actor-critic, on-policy)
Intrinsic modules	Count-based, ICM, RIDE
Compared methods	PPO (no intrinsic) / PPO + fixed- β / PPO + CARE
Environments	6 MiniGrid sparse-reward tasks (5 seeds, 10^6 env steps each)
Metrics	Success rate, episode return, learning speed, stability across seeds

Results – Sample Efficiency



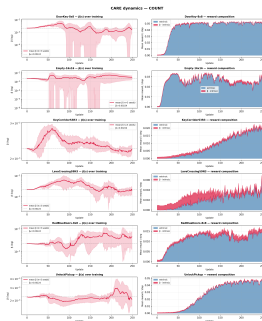
Lower is faster. Aggregated across 6 envs $\times$ 5 seeds (95% bootstrap CI).

Results – Final Performance



Higher is better. Final reward normalized to env-max, last 10 logs averaged.

Analysis – Why CARE Works



Fixed β

Same exploration strength everywhere.

CARE

Per-state β_{ψ} ; shrinks once useful exploration is found.

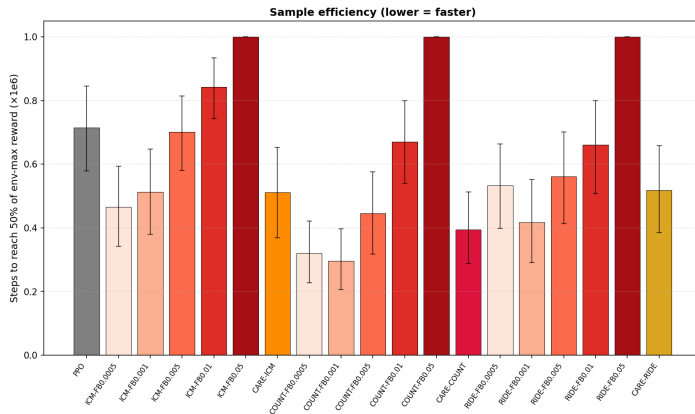
Limitations

- MiniGrid only (discrete, low-dim).
- Extra network + training objective.
- Sensitive to β_0 , λ_{reg} , lr_{ψ} .
- Future: continuous control, image-based.

- Sparse rewards $\Rightarrow$ hard exploration.
- Fixed β = one-size-fits-all.
- **CARE**: adaptive $\beta_\psi(s)$ per state.

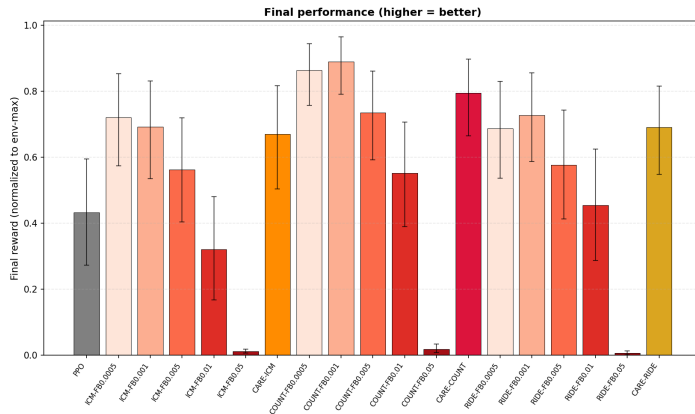
Thank you for listening!

[Backup] Extended β Range – Sample Efficiency



Fixed- β sweep extended to $\beta \in [10^{-8}, 1]$.

[Backup] Extended β Range – Final Performance



Fixed- β sweep extended to $\beta \in [10^{-8}, 1]$.

[Backup] CARE vs Best Fixed β (all 18 cells)

Env	Module	CARE	Best fixed	Best β	Gap
DoorKey-8x8	COUNT	0.973	0.973	0.0005	+0.001
	ICM	0.770	0.973	0.0005	-0.202
	RIDE	0.973	0.973	0.0005	+0.000
Empty-16x16	COUNT	0.959	0.975	0.0005	-0.017
	ICM	0.575	0.975	0.0005	-0.400
	RIDE	0.941	0.974	0.001	-0.033
KeyCorridor	COUNT	0.831	0.828	0.001	+0.003
	ICM	0.000	0.457	0.001	-0.457
	RIDE	0.548	0.703	0.001	-0.155
LavaCrossing	COUNT	0.372	0.530	0.001	-0.158
	ICM	0.077	0.060	0.001	+0.017
	RIDE	0.543	0.512	0.0005	+0.031
RedBlueDoors	COUNT	0.653	0.779	0.0005	-0.126
	ICM	0.273	0.737	0.0005	-0.464
	RIDE	0.593	0.794	0.001	-0.201
UnlockPickup	COUNT	0.936	0.936	0.0005	+0.000
	ICM	0.716	0.937	0.001	-0.221
	RIDE	0.536	0.369	0.0005	+0.167

[Backup] CARE Pseudocode (Algorithm 1)

- ➊ Initialize π_θ , V_ϕ , Beta Network β_ψ , intrinsic module m .
- ➋ Repeat for T steps:
 - ➊ Collect rollout $\{(s_t, a_t, r_t^E, s_{t+1})\}$ with π_θ .
 - ➋ Compute $R_t^{I,m}$ from module m ; standardize to I_t^+ .
 - ➌ Compute $\beta_\psi(s_t)$ and shaped reward $\bar{r}_t = r_t^E + \beta_\psi(s_t) \cdot I_t^+$.
 - ➍ Compute raw extrinsic GAE advantage $\hat{A}_t^E$ and shaped advantage $\hat{A}_t^{\text{shaped}}$.
 - ➎ Update π_θ , V_ϕ with PPO ($K=80$ epochs, normalized $\hat{A}$).
 - ➏ One Adam step on ψ : $\mathcal{L}_\beta = -\mathbb{E}[\hat{I}_z \hat{A}_z^E] + \lambda_{\text{reg}}(\log \beta_\psi - \log \beta_0)^2$.
 - ➐ Clip β_ψ to $[\beta_{\min}, \beta_{\max}]$.

[Backup] Key Hyperparameters

PPO	$K = 80, \epsilon_{\text{clip}} = 0.2, \gamma = 0.99, \lambda_{\text{GAE}} = 0.95$ $\text{lr}_{\text{actor}} = 3 \times 10^{-4}, \text{lr}_{\text{critic}} = 10^{-3}, \text{rollout } T = 4,000$
CARE	$\beta_{\min} = 10^{-4}, \beta_{\max} = 5 \times 10^{-2}$ $\beta_0 = \sqrt{\beta_{\min} \beta_{\max}} \approx 2.24 \times 10^{-3}$ $\lambda_{\text{reg}} = 10^{-3}, \text{lr}_{\psi} = 5 \times 10^{-4}, \text{weight decay } 10^{-6}$ one ψ -step per PPO update, grad-clip L2 = 1.0, no delayed target copy
Fixed-β sweep	$\beta \in \{0.0005, 0.001, 0.005, 0.01, 0.05\}$
Seeds	$\{1, 2, 3, 4, 5\}, 10^6$ env steps per run